

Environmental Compensation Unit

accurate interferometrical measurements in ambient environment

ECU/IDS3010

In order to reduce position inaccuracy due to air-induced variations of the index refraction, attocube supplies an environmental compensation unit (ECU). By locally measuring environmental parameters, an accuracy of typically better ± 1 ppm can be achieved in air. The ECU is plug and-play compatible with all IDS models and can be screw mounted.



IDS ECU

Technical Specifications	Art. No.
art. no.	1010698
dimensions	28 x 61 x 20.5 mm ³
weight	41.5 g
integrated sensors	T, p, rH
interface	RJ12 connector
sensor mount	screw mount
working environment	non condensing
cable length	1,83 m
Measurement Accuracy (Sensors)	
T-sensor	$\pm 0.1^{\circ}\text{C}$ (0..50°C)
p-sensor	± 1 hPa (300..1100 mbar)
rH-sensor	$\pm 2\%$ (10..90%)
typ. accuracy setup	guaranteed ± 1.0 ppm up to 5m; expected ± 0.0 ppm up to 4,2 m

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ZOOM OUT Ver.

BME680 Accuracy

Here's the accuracy of the temperature, humidity and pressure sensors of the BME680:

Sensor	Accuracy
Temperature	+/- 1.0°C
Humidity	+/- 3%
Pressure	+/- 1 hPa

BME680 Operation Range

The following table shows the operation range for the temperature, humidity and pressure sensors for the BME680.

Sensor	Operation Range
Temperature	-40 to 85 °C
Humidity	0 to 100 %
Pressure	300 to 1100 hPa

There's a significant difference in between BME680 accuracy and ECU(from attocube), especially Temperature.

Key features

- Package 3.0 mm x 3.0 mm x 0.93 mm metal lid LGA
- Digital interface I²C (up to 3.4 MHz) and SPI (3 and 4 wire, up to 10 MHz)
- Supply voltage V_{DD} main supply voltage range: 1.71 V to 3.6 V
V_{DDIO} interface voltage range: 1.2 V to 3.6 V
- Current consumption 2.1 µA at 1 Hz humidity and temperature
3.1 µA at 1 Hz pressure and temperature
3.7 µA at 1 Hz humidity, pressure and temperature
0.09–12 mA for p/h/T/gas depending on operation mode
0.15 µA in sleep mode
- Operating range -40–+85 °C, 0–100% r.H., 300–1100 hPa
- Individual humidity, pressure and gas sensors can be independently enabled/disabled
- The product is RoHS compliant, halogen-free, MSL1

Key parameters for gas sensor

- Response time ($\tau_{33-63\%}$) < 1 s (for new sensors)
- Power consumption < 0.1 mA in ultra-low power mode
- Output data processing direct indoor air quality (IAQ) index output

Key parameters for humidity sensor

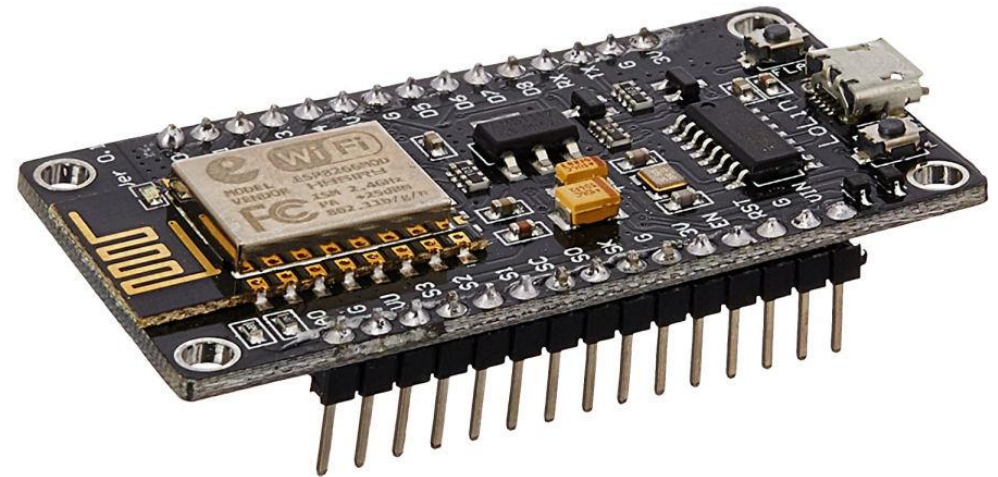
- Response time ($\tau_{0-63\%}$) ~8 s
- Accuracy tolerance ±3% r.H.
- Hysteresis ±1.5% r.H.

Key parameters for pressure sensor

- RMS Noise 0.12 Pa, equiv. to 1.7 cm
- Offset temperature coefficient ±1.3 Pa/K, equiv. to ±10.9 cm at 1 °C temperature change

NODEMCU

- Better version of Arduino (Powerful processor)
- Main Component: ESP8266 with programmable pins and built-in **wifi**.
- Power Via USB
- Low cost (Price range: 2.44 – 5£, eBay price: 3.99£)
- Main Application: Internet of Things (IOT)
- NodeMCU features two hardware UARTs (UART0 and UART1) with baud rates of up to 115200.
- It can communicate at a maximum speed of 4.5 Mbps.
- 80MHZ of clock speed and 4Mb flash memory



Just like we can monitor IDS in real-time, we can easily monitor BME680 readings using NodeMCU via phone or desktop(remotely).

